Tae Hyeon Kweon

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Education

University of Waterloo Ontario, Canada
MASc in Mechanical and Mechatronics Engineering | GPA: 88.5 Sep. 2024 – Present
Advisor: Prof. Soo Jeon
Relevant Coursework: Optimal Control, Adaptive Control, Video Processing, Machine Learning

The Hong Kong Polytechnic University (HKPolyU) Kowloon, Hong Kong
B.Eng. (Honours) in Mechanical Engineering Sep. 2018 – Jun. 2024
Graduated with First Class Honors (Mandatory military service: 2021–2023)
Thesis: *A Mobile Robot with an Active Suspension System for Navigation in Uneven Terrain*

Publications

Tae Hyeon Kweon, Soo Jeon. *Semantic-Aware Active Perception for Next-Best-View Grasp Planning.*
UWSpace Preprint, Under Review, 2026.
`https://uwspace.uwaterloo.ca/items/0a994fe1-b395-4395-a1ca-b0f7c20c2605`

Tae Hyeon Kweon, Soo Jeon. *Learning-Based View Scoring for Next-Best-View Planning Using Vision-Language Models.*
Manuscript in preparation, 2026.

Research & Work Experience

Mechanical Systems Control Lab, UWaterloo Ontario, Canada
Graduate Student Researcher Sep. 2024 – Present
Advisor: Prof. Soo Jeon
– Developed a semantic Next-Best-View framework for grasp planning under occlusion, integrating semantic segmentation, volumetric mapping, and grasp synthesis.
– Investigating verification of vision-language-action (VLA) policies using predictive world models to identify potential failures before execution.

Robotics and Machine Intelligence Lab, HKPolyU Kowloon, Hong Kong
Student Researcher (Undergraduate Research and Innovation Scheme) Sep. 2023 – Jun. 2024
Advisor: Prof. David Navarro-Alarcon
– Designed motion-planning and control strategies for a dual-arm TIAGo++ robot performing bimanual manipulation.

Undergraduate Research Assistant Dec. 2020 – Jun. 2021
– Built a mobile robot capable of autonomous navigation over liquid and muddy terrain.
– Implemented a vision-based localization and navigation system using artificial markers.

Origami Labs (Start-up) Tsuen Wan, Hong Kong
Engineering Intern Jul. 2020 – Sep. 2020
– Integrated an accelerometer-based trigger system for automatic microphone activation.
– Applied signal-processing filters to reduce environmental noise in wearable prototypes.

Selected Projects

End-to-End Visual Grasping with PPO, UWaterloo Jan. 2025 – Apr. 2025
SYDE 673 Video Processing and Analysis (Course Project)
– Developed and trained a PPO-based visuomotor grasping policy using the Barrett WAM arm

in the Genesis simulator.
– Designed a visual reward function based on corner overlap, contact detection, and lift height.
– Built a multi-environment camera system for robust, parallelized vision-based training.

Adaptive Control of a 2-DOF Robotic Arm, UWaterloo Sep. 2024 – Dec. 2024
ME780 Adaptive Control (Course Project)
– Implemented a Model Reference Adaptive Control (MRAC) scheme to adaptively estimate unknown dynamics and ensure stable trajectory tracking.
– Derived Lyapunov-based adaptation laws and validated controller performance in MATLAB/Simulink.

Active Suspension for Navigating Rough Terrain, HKPolyU Sep. 2023 – Jun. 2024
Undergraduate Thesis
– Designed a mobile robot with four independently actuated suspension units driven by stepper motors.
– Integrated IMU feedback to detect terrain pitch and actively adjust body height for stable navigation improving robustness on uneven terrain.

Teaching Experience

ME549/MTE544 Autonomous Mobile Robotics, University of Waterloo
Teaching Assistant Winter 2025, Fall 2025, Winter 2026
– Led lab sessions on localization, control, and planning; held office hours; graded assignments and exams.

Scholarships & Awards

International Master's Award of Excellence (IMAE), UWaterloo	2024 – 2026
Undergraduate Research Program Scholarship, HKPolyU	2023
Dean's Honour List, HKPolyU	2021
Full Entry Scholarship, HKPolyU	2018 – 2024

Extracurricular Activities

Volunteer (Score Tally), MME Graduate Research Symposium, UWaterloo	Nov. 2024
Peer Tutor in AMA1120 (Basic Mathematics II), HKPolyU	Jan. 2024 – May 2024
Mentor, Korean Student Association, HKPolyU	Sep. 2023 – Dec. 2023
Team Leader, STEM Learning Kits for Overseas Students, HKPolyU	Jun. 2020 – Aug. 2020

Skills

Programming:	Python, C++, MATLAB, Shell
Tools:	PyTorch, ROS 1 & 2, OpenCV, Open3D, Git, Docker, LaTeX
Robots:	Franka Panda, Barrett WAM, TIAGO++, TurtleBot
Simulation:	PyBullet, Genesis, Gazebo, Simulink
Hardware:	IMU, RealSense L515/D455, Arduino, Jetson Orin, Raspberry Pi